

Aryan Varshney

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SUMMARY

AI/ML Engineer with hands-on experience building production Generative AI systems — real-time voice pipelines, RAG applications, and computer vision — using Python, LangChain, FAISS, and LLM APIs (LLaMA, Gemini). Strong foundations in DSA and end-to-end ML model development with Scikit-learn.

EXPERIENCE

EduLinkUp

Remote

AI/ML Intern — Generative AI (RAG, LLMs, FAISS, Prompt Engineering)

Jan 2026 – Feb 2026

- Built a Generative AI cybersecurity tool using Gemini 2.5 Flash with **RAG** over a **FAISS vector index** of **1,800+ document pages and 10 web sources**, enabling accurate, grounded responses to cybersecurity queries.

PROJECTS

Vayu – Interactive Face Recognition Bot | Python, OpenCV, DeepFace (Facenet512), FAISS, SQLite, Sarvam AI

- Built a real-time AI booth assistant for IIIT Converge'26 that identified **50+ attendees** with **90% recognition accuracy** via FAISS cosine-similarity search over Facenet512 embeddings, delivering fully personalized spoken responses through a conversational pipeline, with average face recognition latency of **10ms per frame**.
- Developed an end-to-end voice pipeline using **Speech Recognition** (Speech-to-Text → LLaMA 3.3-70B → Text-to-Speech over WebSocket) with streaming audio playback that begins speaking within **0.4 seconds** of the first audio chunk arriving.
- Engineered concurrent execution using Python's **ProcessPoolExecutor** to run face recognition in an isolated process alongside live **Computer Vision (OpenCV)** and a **Silero VAD** audio thread, ensuring smooth, non-blocking operation.

Stock Market Voice | Python, Sarvam AI, Groq, Twilio, Silero VAD

- Stock Market Voice, an AI-powered voice calling system for brokers that schedules and initiates automated calls to clients, delivering personalized daily stock purchase summaries with real-time conversational interaction.
- Architected an end-to-end voice pipeline (Speech-to-Text → LLaMA 3.3-70B → Text-to-Speech) integrated with **Silero VAD** for speech detection, enabling **1–2s response latency** from speech input to spoken reply over calls.
- Leveraged Twilio for scalable call automation and broker APIs for dynamic retrieval of client-specific stock data, supporting multi-user calling and seamless callback-based query handling.

Sleep Debt Predictor (Live Demo) | Python, FastAPI, Scikit-learn, Linear Regression

- Sleep Debt Predictor, a web-based ML system that estimates user sleep debt by analyzing facial fatigue indicators (eye redness, dark circles, yawn frequency) captured via real-time browser video input.
- Trained a Scikit-learn Linear Regression model on a self-collected dataset, achieving **R² score of 0.87**, **MAE of 0.80 hours**, and **RMSE of 1.02** on the test split, enabling accurate and interpretable sleep debt estimation.
- Built an end-to-end pipeline integrating FastAPI backend with Google Gemini API for feature extraction, and implemented Web Speech API for AI-generated personalized voice feedback.

Spy AI | Flutter, FastAPI, Dart, Python, Gemini, PostgreSQL

- Designed and developed an AI-powered full-stack mobile app using Flutter (Dart) frontend and FastAPI (Python) backend that turns a phone into a searchable, lifelong memory by recording and indexing meetings in real time.
- Implemented a background recorder with real-time audio capture, transcription pipeline, structured database storage and **full-text search**; integrated a Gemini 2.5 Flash-powered chatbot to surface summaries, exact quotes, timestamps and context on demand across past conversations.

EDUCATION

Jaypee Institute of Information Technology

Noida, India

B.Tech. in Computer Science and Engineering

2024 – 2028

TECHNICAL SKILLS

Programming Languages: Python, Java, Dart, C, C++

Frameworks: Django, FastAPI, Flask, Flutter, Spring Boot

Databases: PostgreSQL, MySQL, MongoDB

AI/ML: Machine Learning, Scikit-learn, PyTorch, HuggingFace, Embeddings, FAISS, NLP, LangChain, RAG

Tools: GitHub, Docker, REST APIs, WebSockets, n8n, Gemini API, Sarvam AI, Groq, Silero VAD

ACHIEVEMENTS

Keploy Certified

August 2025

- Top 168 of 18,000+ participants (top 1%)** in Keploy API Fellowship — selected through API design, testing, and documentation rounds